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RECOMBINANT INFLUENZA VIRUSES WITH STABILIZED NA

Abstract

Modified influenza virus neuraminidases are described herein that have stabilized NA tetramers which may improve vaccine production efficiency, thus improving the yield of vaccine viruses.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional of U.S. application Ser. No. 18/365,082, filed Aug. 3, 2023, which is a divisional of U.S. application Ser. No.

INCORPORATION BY REFERENCE OF SEQUENCE LISTING

[0003] A Sequence Listing is provided herewith as an xml file, "2356941.xml" created on Aug. 3, 2023, and having a size of 101,767 bytes. The content of the xml file is incorporated by reference herein in its entirety.

BACKGROUND

[0004] Influenza is a major respiratory disease in some mammals including horses and is responsible for substantial morbidity and economic losses each year. In addition, influenza virus infections can cause severe systemic disease in some avian species, leading to death. The segmented nature of the influenza virus genome allows for reassortment of segments during virus replication in cells infected with two or more influenza viruses. The reassortment of segments, combined with genetic mutation and drift, can give rise to a myriad of divergent strains of influenza virus over time. The new strains exhibit antigenic variation in their hemagglutinin (HA) and/or neuraminidase (NA) proteins, and in particular the gene coding for the HA protein has a high rate of variability. The predominant current practice for the prevention of flu is vaccination. Most commonly, inactivated virus vaccines are used. As the influenza HA protein is the major target antigen for the protective immune responses of a host to the virus and is highly variable, the isolation of influenza virus and the identification and characterization of the HA antigen in viruses associated with recent outbreaks is important for vaccine production. Based on prevalence and prediction, a vaccine is designed to stimulate a protective immune response against the predominant and expected influenza virus strains.

[0005] There are four general types of influenza viruses, Type A, Type B, Type C, and Type D, which are defined by the absence of serological cross reactivity between their internal proteins. Influenza Type A viruses are further classified into subtypes based on antigenic and genetic differences of their glycoproteins, the HA and NA proteins. All the known HA and NA subtypes (H1 to H18 and N1 to N11) have been isolated from aquatic birds, which are thought to act as a natural reservoir for influenza.

[0006] It has been suggested that antibodies against NA play important roles in preventing influenza virus infections. However, the current influenza vaccines, which are made by the inactivation of influenza viruses grown in eggs and purification of the virus antigen, are not able to elicit anti-NA antibodies efficiently.

SUMMARY

[0007] One of the causes of the low production of anti-NA antibodies is attributed to the structural instability of the NA protein, which works as a homo-tetramer. The NA tetramer may be disrupted during the antigen purification process. Therefore, the amount of NA contained in current vaccines is insufficient to elicit the production of anti-NA antibodies.

[0008] The present disclosure relates to influenza viruses with certain residue(s) or modifications in the NA protein that stabilize its natural homotetramer structure, and methods of making and using that virus, e.g., in a vaccine. In one embodiment, an isolated recombinant influenza virus comprising a neuraminidase (NA) viral segment encoding a NA monomer that forms virions having stabilized NA tetramers is provided. In one embodiment, the recombinant influenza virus has a modified NA stalk relative to a parental NA that results in stabilized tetramers. In one embodiment, the modified NA stalk has a deletion. In one embodiment, the modified NA stalk has an insertion relative to a parental NA. In one embodiment, the modified NA stalk has at least one amino acid substitution relative to a parental NA. In one embodiment, at least one substitution in the modified NA stalk is a cysteine substitution. In one embodiment, the modified NA stalk has at least two substitutions relative to a parental NA. In one embodiment, the NA has a cysteine at position 48 relative to the numbering of N1. In one embodiment, the NA has a cysteine at position 50 relative to the numbering of N1. In one embodiment, the NA has a cysteine at position 48 and position 50 relative to the numbering of N1. In one embodiment, the NA stalk is modified within residues 1 to 10 from the C-terminus of the transmembrane domain relative to a parental NA. In one embodiment, the NA stalk is modified within residues 10 to 20 from the C-terminus of the transmembrane domain. In one embodiment, the NA stalk is modified within residues 20 to 30 from the C-terminus of the transmembrane domain. In one embodiment, the NA stalk is modified within residues 30 to 50 from the C-terminus of the transmembrane domain. In one embodiment, the stalk domain begins at the first residue after the last residue in the transmembrane domain of NA up to the conserved cysteine, or one to two residues before the conserved cysteine, in the head region of NA that forms a disulfide bond (see Blumenkrantz et al., *J. Virol.*, 87:10539 (2013)). In one embodiment, the insertion in the NA stalk is an insertion of 1, 2, 3, 4 or 5, or 5 to 10, or 10 to 20, or more residues. In one embodiment, the deletion in the NA stalk is a deletion of 1, 2, 3, 4 or 5, or 5 to 10, or 10 to 20, or more

residues. In one embodiment, a NA that has a cysteine is the stalk region is modified to include a second or third cysteine, e.g., within 1 to 10 or 1 to 5 residues of the other cysteine(s). In one embodiment, a NA has at least two cysteines in the stalk region that are within 1 to 2, 2 to 3, 3 to 4 or 5 up to residues of each other. In one embodiment, a NA has at least two cysteines in the stalk region, one of which is within 5 residues, either N-terminal or C-terminal, of residue 48 in the NA, and the other of which is within 5 residues, either N-terminal or C-terminal, of residue 50 in the NA. In one embodiment, the cysteines are adjacent to each other, e.g., at residues 46 and 47, residues 46, 47 and 48, residues 47 and 48, residues 47, 48 and 49, residues 48 and 49, residues 48, 49, and 50, residues 49 and 50, or residues 49, 50 and 51, and in one embodiment both are in the stalk region. In one embodiment, the cysteines are 2 residues apart and in one embodiment both are in the stalk region, e.g., the cysteines are at residues 46 and 48, 47 and 49, 48 and 50, 49 and 51, 50 and 52, 51 and 53, and the like. In one embodiment, the cysteines are 3 residues apart, e.g., the cysteines are at residues 45 and 48, 46 and 49, 47 and 50, 48 and 51, 49 and 52, 50 and 53, 51 and 54, and the like and in one embodiment both are in the stalk region. In one embodiment, the cysteines are 4 residues apart, e.g., the cysteines are at residues 44 and 48, 45 and 49, 46 and 50, 47 and 51, 48 and 52, 49 and 53, 50 and 54, 51 and 55, and the like and in one embodiment both are in the stalk region. In one embodiment, the stalk region has no more than 10, 9, 8, 7, 6, 5, 4, 3, 2 or 1 cysteine residues. In one embodiment, the cysteine(s) in the stalk region are in the N-terminal 30 residues of the stalk region. In one embodiment, the cysteine(s) in the stalk region are in the N-terminal 20 residues of the stalk region.

[0009] As described herein, a highly proliferative recombinant influenza viruses expressing structurally stabilized neuraminidase tetramers was prepared. In one embodiment, recombinant influenza viruses containing either NA-48C or NA-50C or both express stabilized NA tetramers and replicate efficiently. Thus, amino acid mutations in influenza NA that enable the generation of highly proliferative recombinant influenza viruses expressing structurally stabilized NA tetramers can be used to generate influenza vaccine strains that can be used to produce influenza vaccines containing a greater amount of NA antigen. Influenza vaccine strains containing a greater amount of NA antigen that can elicit anti-NA antibodies efficiently.

[0010] In one embodiment, a vaccine comprising an effective amount of the recombinant influenza virus or a portion thereof is provided. In one embodiment, the vaccine is a whole virus vaccine. In one embodiment, the vaccine is a split virus vaccine. In one embodiment, the vaccine is a subunit vaccine. In one embodiment, the vaccine further comprises an adjuvant. In one embodiment, the vaccine further comprises a pharmaceutically acceptable carrier. In one embodiment, the carrier is suitable for intranasal or intramuscular administration. In one embodiment, the vaccine further comprises at least one other influenza virus isolate.

[0011] In one embodiment, a method of preparing influenza virus having stabilized NA tetramers is provided. The method includes contacting a cell with one or more vectors comprising nucleic acid for an influenza virus NA segment encoding, for example, at least one cysteine in the stalk region, nucleic acid for an influenza virus PA segment, nucleic acid for an influenza virus PB1 segment, nucleic acid for an influenza virus PB2 segment, nucleic acid for an influenza virus NP segment, nucleic acid for an influenza virus NS segment, nucleic acid for an influenza virus M segment, and nucleic acid for an influenza virus HA segment, in an amount effective to produce influenza virus having stabilized NA tetramers. In one embodiment, the NA is N1, N2, N3 or N5. In one embodiment, the NA is N4, N6, N7, N8 or N9. In one embodiment, the cell is a mammalian cell. In one embodiment, the cell is a Vero cell, 293T cell or MDCK cell.

[0012] In one embodiment, a method of making an influenza vaccine is provided. The method includes combining the recombinant virus with an adjuvant or treating the virus with an agent that inactivates the virus. In one embodiment, the method further comprises aliquoting a selected dose of the virus into a receptacle. In one embodiment, the adjuvant comprises immunostimulatory DNA sequences, bacterium-derived components, aluminum salt (alum) or squalene oil-in-water emulsion systems such as MF59 and AS03. In one embodiment, the agent chemically inactivates the virus. In one embodiment, the agent comprises formalin or beta-propiolactone. In one embodiment, the agent comprises a detergent, e.g., a non-ionic detergent, a cationic detergent or an anionic detergent. In one embodiment, the detergent comprises CTAB, Triton, SDS, Neodol 23-6, or sodium desoxycholate. In one embodiment, the method further comprises separating HA and NA from other viral components, e.g., using centrifugation and collection of soluble molecules.

[0013] In one embodiment, a method of preparing influenza virus is provided that includes contacting cells with the recombinant virus in an amount effective to yield progeny virus. In one embodiment, the virus is contacted with an avian egg. In one embodiment, the cells are mammalian cells. In one embodiment, the HA of the virus is H1, H3, H5 or H7.

[0014] Also provided is a method of preparing stabilized NA tetramers, comprising: contacting a cell with one or more vectors comprising nucleic acid for an influenza virus NA segment encoding at least one cysteine in the stalk region and nucleic acid for an influenza virus HA. In one embodiment, the method further comprises isolating NA and HA from the cell. In one embodiment, the cell is an insect cell. [0015] Further provided is a method of immunizing an avian or a mammal, comprising: administering to the avian or the mammal a composition having an effective amount of the recombinant virus. In one embodiment, the composition comprises at least one other different influenza virus. In one embodiment, the mammal is a human. In one embodiment, the composition is administered intranasally or via injection.

Description

BRIEF DESCRIPTION OF FIGURES

[0016] FIG. 1. Western blot to detect NA in hCK cells infected with influenza viruses.
[0017] FIG. 2. Amino acid sequence of the NA of A/Yokohama/2017/2003 (SEQ ID NO:1).
[0018] FIG. 3. Amino acid sequence for the NA of A/Saitama/103/2014 (SEQ ID NO:2).
[0019] FIG. 4. Amino acid sequences for NA of mutant of A/Yokohama/2017/2003 (SEQ ID NO:3).
[0020] FIG. 5. Exemplary NA sequences for N3, N4, N6, N7, N8, and N9 (SEQ ID Nos. 4-9). The stalk region is indicated by red font, underlining.
[0021] FIG. 6. Exemplary viral backbone sequences (SEQ ID Nos. 10-14).
[0022] FIG. 7. Exemplary NA sequences (SEQ ID Nos. 15-18). The stalk region is indicated by red font, underlining.
[0023] FIG. 8. Exemplary NA nucleic acid sequences (SEQ ID Nos 19-29).
[0024] FIG. 9. Exemplary influenza B virus NA sequences (SEQ ID Nos. 45-50). In one embodiment, the stalk region in the NA of influenza B virus may be from residue 38 to 86.
[0025] FIG. 10. Virus growth curves. 6+2 reassortant influenza viruses containing the HA and NA viral segments from A/Singapore/GP1908/2015 (H1N1) pdm09 in the backbone of high-yield A/Puerto Rico/8/1934 (H1N1) were prepared using reverse genetics and propagation in hCK cells at 37° C. Mutant viruses with either the NA-T48C or NA-N50C mutation or both mutations were generated. hCK cells were infected with these viruses at a MOI (multiplicity of infection) of 0.001. The supernatants were collected at the indicated times post-infection and virus titers were determined by means of plaque assay in hCK cells.

DETAILED DESCRIPTION

Definitions

[0026] As used herein, the term “isolated” refers to in vitro preparation and/or isolation of a nucleic acid molecule, e.g., vector or plasmid, peptide or polypeptide (protein), or virus of the invention, so that it is not associated with in vivo substances, or is substantially purified from in vitro substances. An isolated virus preparation is generally obtained by in vitro culture and propagation, and/or via passage in eggs, and is substantially free from other infectious agents.
[0027] As used herein, “substantially purified” means the object species is the predominant species, e.g., on a molar basis it is more abundant than any other individual species in a composition, and preferably is at least about 80% of the species present, and optionally 90% or greater, e.g., 95%, 98%, 99% or more, of the species present in the composition.
[0028] As used herein, “substantially free” means below the level of detection for a particular infectious agent using standard detection methods for that agent.
[0029] A “recombinant” virus is one which has been manipulated in vitro, e.g., using recombinant DNA techniques, to introduce changes to the viral genome. Reassortant viruses can be prepared by recombinant or nonrecombinant techniques.
[0030] As used herein, the term “recombinant nucleic acid” or “recombinant DNA sequence or segment” refers to a nucleic acid, e.g., to DNA, that has been derived or isolated from a source, that may be subsequently chemically altered in vitro, so that its sequence is not naturally occurring, or corresponds to naturally occurring sequences that are not positioned as they would be positioned in the native genome. An example of DNA “derived” from a source, would be a DNA sequence that is identified as a useful fragment, and which is then chemically synthesized in essentially pure form. An example of such DNA “isolated” from a source would be a useful DNA sequence that is excised or removed from said source by chemical means, e.g., by the use of restriction endonucleases, so that it can be further manipulated, e.g., amplified, for use in the disclosure, by the methodology of genetic engineering.

[0031] As used herein, a “heterologous” influenza virus gene or viral segment is from an influenza virus source that is different than a majority of the other influenza viral genes or viral segments in a recombinant, e.g., reassortant, influenza virus.

[0032] The terms “isolated polypeptide”, “isolated peptide” or “isolated protein” include a polypeptide, peptide or protein encoded by cDNA or recombinant RNA including one of synthetic origin, or some combination thereof.

[0033] The term “recombinant protein” or “recombinant polypeptide” as used herein refers to a protein molecule expressed from a recombinant DNA molecule. In contrast, the term “native protein” is used herein to indicate a protein isolated from a naturally occurring (i.e., a nonrecombinant) source. Molecular biological techniques may be used to produce a recombinant form of a protein with identical properties as compared to the native form of the protein.

[0034] Methods of alignment of sequences for comparison are well known in the art. Thus, the determination of percent identity between any two sequences can be accomplished using a mathematical algorithm.

[0035] Computer implementations of these mathematical algorithms can be utilized for comparison of sequences to determine sequence identity. Alignments using these programs can be performed using the default parameters. Software for performing BLAST analyses is publicly available through the National Center for Biotechnology Information (<http://www.ncbi.nlm.nih.gov/>). The algorithm may involve first identifying high scoring sequence pairs (HSPs) by identifying short words of length W in the query sequence, which either match or satisfy some positive-valued threshold score T when aligned with a word of the same length in a database sequence. T is referred to as the neighborhood word score threshold. These initial neighborhood word hits act as seeds for initiating searches to find longer HSPs containing them. The word hits are then extended in both directions along each sequence for as far as the cumulative alignment score can be increased. Cumulative scores are calculated using, for nucleotide sequences, the parameters M (reward score for a pair of matching residues; always >0) and N (penalty score for mismatching residues; always <0). For amino acid sequences, a scoring matrix is used to calculate the cumulative score. Extension of the word hits in each direction are halted when the cumulative alignment score falls off by the quantity X from its maximum achieved value, the cumulative score goes to zero or below due to the accumulation of one or more negative-scoring residue alignments, or the end of either sequence is reached.

[0036] In addition to calculating percent sequence identity, the BLAST algorithm may also perform a statistical analysis of the similarity between two sequences. One measure of similarity provided by the BLAST algorithm may be the smallest sum probability ($P(N)$), which provides an indication of the probability by which a match between two nucleotide or amino acid sequences would occur by chance. For example, a test nucleic acid sequence is considered similar to a reference sequence if the smallest sum probability in a comparison of the test nucleic acid sequence to the reference nucleic acid sequence is less than about 0.1, more preferably less than about 0.01, and most preferably less than about 0.001.

[0037] The BLASTN program (for nucleotide sequences) may use as defaults a wordlength (W) of 11, an expectation (E) of 10, a cutoff of 100, $M=5$, $N=-4$, and a comparison of both strands. For amino acid sequences, the BLASTP program may use as defaults a wordlength (W) of 3, an expectation (E) of 10, and the BLOSUM62 scoring matrix. See <http://www.ncbi.nlm.nih.gov>. Alignment may also be performed manually by inspection.

[0038] For sequence comparison, typically one sequence acts as a reference sequence to which test sequences are compared. When using a sequence comparison algorithm, test and reference sequences are input into a computer, subsequence coordinates are designated if necessary, and sequence algorithm program parameters are designated. The sequence comparison algorithm then calculates the percent sequence identity for the test sequence(s) relative to the reference sequence, based on the designated program parameters.

Influenza Virus Structure and Propagation

[0039] Influenza A viruses possess a genome of eight single-stranded negative-sense viral RNAs (vRNAs) that encode at least ten proteins. The influenza virus life cycle begins with binding of the hemagglutinin (HA) to sialic acid-containing receptors on the surface of the host cell, followed by receptor-mediated endocytosis. The low pH in late endosomes triggers a conformational shift in the HA, thereby exposing the N-terminus of the HA2 subunit (the so-called fusion peptide). The fusion peptide initiates the fusion of the viral and endosomal membrane, and the matrix protein (M1) and RNP complexes are released into the cytoplasm. RNPs consist of the nucleoprotein (NP), which encapsidates vRNA, and the viral polymerase complex, which is formed by the PA, PB1, and PB2 proteins. RNPs are transported into the nucleus, where transcription and replication take place. The RNA polymerase complex catalyzes three different reactions: synthesis of an

mRNA with a 5' cap and 3' polyA structure, of a full-length complementary RNA (cRNA), and of genomic vRNA using the cRNA as a template. Newly synthesized vRNAs, NP, and polymerase proteins are then assembled into RNPs, exported from the nucleus, and transported to the plasma membrane, where budding of progeny virus particles occurs. The neuraminidase (NA) protein plays a crucial role late in infection by removing sialic acid from sialyloligosaccharides, thus releasing newly assembled virions from the cell surface and preventing the self aggregation of virus particles. Although virus assembly involves protein-protein and protein-vRNA interactions, the nature of these interactions is largely unknown.

[0040] Although influenza B and C viruses are structurally and functionally similar to influenza A virus, there are some differences. For example, influenza B virus does not have a M2 protein with ion channel activity but has BM2 and has a viral segment with both NA and NB sequences. Influenza C virus has only seven viral segments.

Cells That can be Used to Produce Virus

[0041] Any cell, e.g., any avian or mammalian cell, such as avian eggs, a human, e.g., 293T or PER.C6® cells, or canine, bovine, equine, feline, swine, ovine, rodent, for instance mink, e.g., MvLu1 cells, or hamster, e.g., CHO cells, or non-human primate, e.g., Vero cells, including mutant cells, which supports efficient replication of influenza virus can be employed to isolate and/or propagate influenza viruses. Isolated viruses can be used to prepare a reassortant virus. In one embodiment, host cells for vaccine production are continuous mammalian or avian cell lines or cell strains. A complete characterization of the cells to be used, may be conducted so that appropriate tests for purity of the final product can be included. Data that can be used for the characterization of a cell includes (a) information on its origin, derivation, and passage history; (b) information on its growth and morphological characteristics; (c) results of tests of adventitious agents; (d) distinguishing features, such as biochemical, immunological, and cytogenetic patterns which allow the cells to be clearly recognized among other cell lines; and (e) results of tests for tumorigenicity. In one embodiment, the passage level, or population doubling, of the host cell used is as low as possible.

[0042] In one embodiment, the cells are WHO certified, or certifiable, continuous cell lines. The requirements for certifying such cell lines include characterization with respect to at least one of genealogy, growth characteristics, immunological markers, virus susceptibility tumorigenicity and storage conditions, as well as by testing in animals, eggs, and cell culture. Such characterization is used to confirm that the cells are free from detectable adventitious agents. In some countries, karyology may also be required. In addition, tumorigenicity may be tested in cells that are at the same passage level as those used for vaccine production. The virus may be purified by a process that has been shown to give consistent results, before vaccine production (see, e.g., World Health Organization, 1982).

[0043] Virus produced by the host cell may be highly purified prior to vaccine or gene therapy formulation. Generally, the purification procedures result in extensive removal of cellular DNA and other cellular components, and adventitious agents. Procedures that extensively degrade or denature DNA may also be used.

Influenza Vaccines

[0044] A vaccine includes an isolated recombinant influenza virus of the invention, and optionally one or more other isolated viruses including other isolated influenza viruses, one or more immunogenic proteins or glycoproteins of one or more isolated influenza viruses or one or more other pathogens, e.g., an immunogenic protein from one or more bacteria, non-influenza viruses, yeast or fungi, or isolated nucleic acid encoding one or more viral proteins (e.g., DNA vaccines) including one or more immunogenic proteins of the isolated influenza virus of the invention. In one embodiment, the influenza viruses of the invention may be vaccine vectors for influenza virus or other pathogens.

[0045] A complete virion vaccine may be concentrated by ultrafiltration and then purified by zonal centrifugation or by chromatography. Viruses other than the virus of the invention, such as those included in a multivalent vaccine, may be inactivated before or after purification using formalin or beta-propiolactone, for instance.

[0046] A subunit vaccine comprises purified glycoproteins. Such a vaccine may be prepared as follows: using viral suspensions fragmented by treatment with detergent, the surface antigens are purified, by ultracentrifugation for example. The subunit vaccines thus contain mainly HA protein, and also NA. The detergent used may be cationic detergent for example, such as hexadecyl trimethyl ammonium bromide (Bachmeyer, 1975), an anionic detergent such as ammonium deoxycholate (layer & Webster, 1976); or a nonionic detergent such as that commercialized under the name TRITON X100. The hemagglutinin may also be isolated after treatment of the virions with a protease such as bromelain, and then purified. The subunit

vaccine may be combined with an attenuated virus of the invention in a multivalent vaccine.

[0047] A split vaccine comprises virions which have been subjected to treatment with agents that dissolve lipids. A split vaccine can be prepared as follows: an aqueous suspension of the purified virus obtained as above, inactivated or not, is treated, under stirring, by lipid solvents such as ethyl ether or chloroform, associated with detergents. The dissolution of the viral envelope lipids results in fragmentation of the viral particles. The aqueous phase is recuperated containing the split vaccine, constituted mainly of hemagglutinin and neuraminidase with their original lipid environment removed, and the core or its degradation products. Then the residual infectious particles are inactivated if this has not already been done. The split vaccine may be combined with an attenuated virus of the invention in a multivalent vaccine.

[0048] Inactivated Vaccines. Inactivated influenza virus vaccines are provided by inactivating replicated virus using known methods, such as, but not limited to, formalin or β -propiolactone treatment. Inactivated vaccine types that can be used in the invention can include whole-virus (WV) vaccines or subvirion (SV) (split) vaccines. The WV vaccine contains intact, inactivated virus, while the SV vaccine contains purified virus disrupted with detergents that solubilize the lipid-containing viral envelope, followed by chemical inactivation of residual virus.

[0049] In addition, vaccines that can be used include those containing the isolated HA and NA surface proteins, which are referred to as surface antigen or subunit vaccines.

[0050] Live Attenuated Virus Vaccines. Live, attenuated influenza virus vaccines, such as those including a recombinant virus of the invention can be used for preventing or treating influenza virus infection.

Attenuation may be achieved in a single step by transfer of attenuated genes from an attenuated donor virus to a replicated isolate or reassorted virus according to known methods. Since resistance to influenza A virus is mediated primarily by the development of an immune response to the HA and/or NA glycoproteins, the genes coding for these surface antigens come from the reassorted viruses or clinical isolates. The attenuated genes are derived from an attenuated parent. In this approach, genes that confer attenuation generally do not code for the HA and NA glycoproteins.

[0051] Viruses (donor influenza viruses) are available that are capable of reproducibly attenuating influenza viruses, e.g., a cold adapted (ca) donor virus can be used for attenuated vaccine production. Live, attenuated reassortant virus vaccines can be generated by mating the ca donor virus with a virulent replicated virus. Reassortant progeny are then selected at 25° C. (restrictive for replication of virulent virus), in the presence of an appropriate antiserum, which inhibits replication of the viruses bearing the surface antigens of the attenuated ca donor virus. Useful reassortants are: (a) infectious, (b) attenuated for seronegative non-adult mammals and immunologically primed adult mammals, (c) immunogenic and (d) genetically stable. The immunogenicity of the ca reassortants parallels their level of replication. Thus, the acquisition of the six transferable genes of the ca donor virus by new wild-type viruses has reproducibly attenuated these viruses for use in vaccinating susceptible mammals both adults and non-adult.

[0052] Other attenuating mutations can be introduced into influenza virus genes by site-directed mutagenesis to rescue infectious viruses bearing these mutant genes. Attenuating mutations can be introduced into non-coding regions of the genome, as well as into coding regions. Such attenuating mutations can also be introduced into genes other than the HA or NA, e.g., the PB2 polymerase gene. Thus, new donor viruses can also be generated bearing attenuating mutations introduced by site-directed mutagenesis, and such new donor viruses can be used in the production of live attenuated reassortants vaccine candidates in a manner analogous to that described above for the ca donor virus. Similarly, other known and suitable attenuated donor strains can be reassorted with influenza virus to obtain attenuated vaccines suitable for use in the vaccination of mammals.

[0053] In one embodiment, such attenuated viruses maintain the genes from the virus that encode antigenic determinants substantially similar to those of the original clinical isolates. This is because the purpose of the attenuated vaccine is to provide substantially the same antigenicity as the original clinical isolate of the virus, while at the same time lacking pathogenicity to the degree that the vaccine causes minimal chance of inducing a serious disease condition in the vaccinated mammal.

[0054] The viruses in a multivalent vaccine can thus be attenuated or inactivated, formulated and administered, according to known methods, as a vaccine to induce an immune response in an animal, e.g., a mammal. Methods are well-known in the art for determining whether such attenuated or inactivated vaccines have maintained similar antigenicity to that of the clinical isolate or high growth strain derived therefrom. Such known methods include the use of antisera or antibodies to eliminate viruses expressing antigenic determinants of the donor virus; chemical selection (e.g., amantadine or rimantidine); HA and NA activity

and inhibition; and nucleic acid screening (such as probe hybridization or PCR) to confirm that donor genes encoding the antigenic determinants (e.g., HA or NA genes) are not present in the attenuated viruses.

Pharmaceutical Compositions

[0055] Pharmaceutical compositions of the present disclosure, suitable for inoculation, e.g., nasal, parenteral or oral administration, comprise one or more influenza virus isolates, e.g., one or more attenuated or inactivated influenza viruses, a subunit thereof, isolated protein(s) thereof, and/or isolated nucleic acid encoding one or more proteins thereof, optionally further comprising sterile aqueous or non-aqueous solutions, suspensions, and emulsions. The compositions can further comprise auxiliary agents or excipients, as known in the art. The composition of the invention is generally presented in the form of individual doses (unit doses).

[0056] Conventional vaccines generally contain about 0.1 to 200 µg, e.g., 30 to 100 µg, 0.1 to 2 µg, 0.5 to 5 µg, 1 to 10 µg, 10 µg to 20 µg, 15 µg to 30 µg, or 10 to 30 µg, of HA from each of the strains entering into their composition. The vaccine forming the main constituent of the vaccine composition of the invention may comprise a single influenza virus, or a combination of influenza viruses, for example, at least two or three influenza viruses, including one or more reassortant(s).

[0057] Preparations for parenteral administration include sterile aqueous or non-aqueous solutions, suspensions, and/or emulsions, which may contain auxiliary agents or excipients known in the art. Examples of non-aqueous solvents are propylene glycol, polyethylene glycol, vegetable oils such as olive oil, and injectable organic esters such as ethyl oleate. Carriers or occlusive dressings can be used to increase skin permeability and enhance antigen absorption. Liquid dosage forms for oral administration may generally comprise a liposome solution containing the liquid dosage form. Suitable forms for suspending liposomes include emulsions, suspensions, solutions, syrups, and elixirs containing inert diluents commonly used in the art, such as purified water. Besides the inert diluents, such compositions can also include adjuvants, wetting agents, emulsifying and suspending agents, or sweetening, flavoring, or perfuming agents.

[0058] When a composition of the present disclosure is used for administration to an individual, it can further comprise salts, buffers, adjuvants, or other substances which are desirable for improving the efficacy of the composition. For vaccines, adjuvants, substances which can augment a specific immune response, can be used. Normally, the adjuvant and the composition are mixed prior to presentation to the immune system, or presented separately, but into the same site of the organism being immunized.

[0059] Heterogeneity in a vaccine may be provided by mixing replicated influenza viruses for at least two influenza virus strains, such as 2-20 strains or any range or value therein. Vaccines can be provided for variations in a single strain of an influenza virus, using techniques known in the art.

[0060] A pharmaceutical composition according to the present disclosure may further or additionally comprise at least one chemotherapeutic compound, for example, for gene therapy, immunosuppressants, anti-inflammatory agents or immune enhancers, and for vaccines, chemotherapeutics including, but not limited to, gamma globulin, amantadine, guanidine, hydroxybenzimidazole, interferon-α, interferon-β, interferon-γ, tumor necrosis factor-alpha, thiosemicarbarzones, methisazone, rifampin, ribavirin, a pyrimidine analog, a purine analog, foscarnet, phosphonoacetic acid, acyclovir, dideoxynucleosides, a protease inhibitor, or ganciclovir.

[0061] The composition can also contain variable but small quantities of endotoxin-free formaldehyde, and preservatives, which have been found safe and not contributing to undesirable effects in the organism to which the composition is administered.

Pharmaceutical Purposes

[0062] The administration of the composition (or the antisera that it elicits) may be for either a “prophylactic” or “therapeutic” purpose. When provided prophylactically, the compositions of the invention which are vaccines are provided before any symptom or clinical sign of a pathogen infection becomes manifest. The prophylactic administration of the composition serves to prevent or attenuate any subsequent infection. When provided prophylactically, the gene therapy compositions of the invention, are provided before any symptom or clinical sign of a disease becomes manifest. The prophylactic administration of the composition serves to prevent or attenuate one or more symptoms or clinical signs associated with the disease.

[0063] When provided therapeutically, a viral vaccine is provided upon the detection of a symptom or clinical sign of actual infection. The therapeutic administration of the compound(s) serves to attenuate any actual infection. When provided therapeutically, a gene therapy composition is provided upon the detection of a symptom or clinical sign of the disease. The therapeutic administration of the compound(s) serves to attenuate a symptom or clinical sign of that disease.

[0064] Thus, a vaccine composition of the present disclosure may be provided either before the onset of infection (so as to prevent or attenuate an anticipated infection) or after the initiation of an actual infection. Similarly, for gene therapy, the composition may be provided before any symptom or clinical sign of a disorder or disease is manifested or after one or more symptoms are detected.

[0065] A composition is said to be “pharmacologically acceptable” if its administration can be tolerated by a recipient mammal. Such an agent is said to be administered in a “therapeutically effective amount” if the amount administered is physiologically significant. A composition of the present invention is physiologically significant if its presence results in a detectable change in the physiology of a recipient patient, e.g., enhances at least one primary or secondary humoral or cellular immune response against at least one strain of an infectious influenza virus.

[0066] The “protection” provided need not be absolute, i.e., the influenza infection need not be totally prevented or eradicated, if there is a statistically significant improvement compared with a control population or set of mammals. Protection may be limited to mitigating the severity or rapidity of onset of symptoms or clinical signs of the influenza virus infection.

Pharmaceutical Administration

[0067] A composition of the present disclosure may confer resistance to one or more pathogens, e.g., one or more influenza virus strains, by either passive immunization or active immunization. In active immunization, an attenuated live vaccine composition is administered prophylactically to a host (e.g., a mammal), and the host's immune response to the administration protects against infection and/or disease. For passive immunization, the elicited antisera can be recovered and administered to a recipient suspected of having an infection caused by at least one influenza virus strain. A gene therapy composition may yield prophylactic or therapeutic levels of the desired gene product by active immunization.

[0068] In one embodiment, the vaccine is provided to a mammalian female (at or prior to pregnancy or parturition), under conditions of time and amount sufficient to cause the production of an immune response which serves to protect both the female and the fetus or newborn (via passive incorporation of the antibodies across the placenta or in the mother's milk).

[0069] The present disclosure thus includes methods for preventing or attenuating a disorder or disease, e.g., an infection by at least one strain of pathogen such as an influenza virus. As used herein, a vaccine is said to prevent or attenuate a disease if its administration results either in the total or partial attenuation (i.e., suppression) of a clinical sign or condition of the disease, or in the total or partial immunity of the individual to the disease. As used herein, a gene therapy composition is said to prevent or attenuate a disease if its administration results either in the total or partial attenuation (i.e., suppression) of a clinical sign or condition of the disease, or in the total or partial immunity of the individual to the disease.

[0070] A composition having at least one influenza virus of the present disclosure, including one which is attenuated and one or more other isolated viruses, one or more isolated viral proteins thereof, one or more isolated nucleic acid molecules encoding one or more viral proteins thereof, or a combination thereof, may be administered by any means that achieve the intended purposes.

[0071] For example, administration of such a composition may be by various parenteral routes such as subcutaneous, intravenous, intradermal, intramuscular, intraperitoneal, intranasal, oral or transdermal routes. Parenteral administration can be accomplished by bolus injection or by gradual perfusion over time.

[0072] A typical regimen for preventing, suppressing, or treating an influenza virus related pathology, comprises administration of an effective amount of a vaccine composition as described herein, administered as a single treatment, or repeated as enhancing or booster dosages, over a period up to and including between one week and about 24 months, or any range or value therein.

[0073] According to the present disclosure, an “effective amount” of a composition is one that is sufficient to achieve a desired effect. It is understood that the effective dosage may be dependent upon the species, age, sex, health, and weight of the recipient, kind of concurrent treatment, if any, frequency of treatment, and the nature of the effect wanted. The ranges of effective doses provided below are not intended to limit the invention and represent dose ranges.

[0074] The dosage of a live, attenuated or killed virus vaccine for an animal such as a mammalian adult organism may be from about 10^{sup.2}-10^{sup.20}, e.g., 10^{sup.3}-10^{sup.14}, 10^{sup.3}-10^{sup.12}, 10^{sup.2}-10^{sup.10}, 10^{sup.5}-10^{sup.11}, 10^{sup.6}-10^{sup.15}, 10^{sup.2}-10^{sup.10}, or 10^{sup.15}-10^{sup.20} plaque forming units (PFU)/kg, about 10^{sup.2}-10^{sup.10}, 10^{sup.3}-10^{sup.12}, 10^{sup.4}-10^{sup.10}, 10^{sup.5}-10^{sup.11}, 10^{sup.6}-10^{sup.14}, 10^{sup.5}-10^{sup.12}, or 10^{sup.14}-10^{sup.20} plaque forming units (PFU)/kg, or any range or value therein. The dose of one viral isolate vaccine, e.g., in an inactivated vaccine, may range

from about 0.1 to 1000, e.g., 0.1 to 10 μg , 1 to 20 μg , 30 to 100 μg , 10 to 50 μg , 50 to 200 μg , or 150 to 300 μg , of HA protein. However, the dosage should be a safe and effective amount as determined by conventional methods, using existing vaccines as a starting point.

[0075] The dosage of immunoreactive HA in each dose of replicated virus vaccine may be standardized to contain a suitable amount, e.g., 0.1 μg to 1 μg , 0.5 μg to 5 μg , 1 μg to 10 μg , 10 μg to 20 μg , 15 μg to 30 μg , or 30 μg to 100 μg or any range or value therein, or the amount recommended by government agencies or recognized professional organizations. The quantity of NA can also be standardized.

[0076] The dosage of immunoreactive HA in each dose of replicated virus vaccine can be standardized to contain a suitable amount, e.g., 1-50 μg or any range or value therein, or the amount recommended by the U.S. Public Health Service (PHS), which is usually 15 μg , per component for older children >3 years of age, and 7.5 μg per component for children <3 years of age. The quantity of NA can also be standardized, however, this glycoprotein can be labile during the processor purification and storage (Kendal et al., 1980; Kerr et al., 1975). Each 0.5-ml dose of vaccine may contain approximately 0.1 to 0.5 billion viral particles, 0.5 to 2 billion viral particles, 1 to 50 billion virus particles, 1 to 10 billion viral particles, 20 to 40 billion viral particles, 1 to 5 billion viral particles, or 40 to 80 billion viral particles.

Exemplary Viruses and Vaccine Formulations

[0077] The present disclosure provides a method that stabilizes the NA tetrameric structure of influenza virus, recombinant viruses having the stabilized NA, and methods of using that virus. For example, NA-48C and/or NA-50C mutations in influenza generate highly proliferative recombinant influenza viruses expressing structurally stabilized NA tetramers that can be used to generate influenza vaccine strains containing a greater amount of NA antigen, e.g., so as to elicit an effective immune response. The influenza vaccines can comprise live attenuated viruses or an inactivated (killed) preparation, e.g., whole virus, subunit or split virus preparation, or exogenously expressed NA and HA. There are three types of inactivated vaccines: whole virus vaccines, split virus vaccines (e.g., disrupted by a detergent), and subunit vaccines (i.e., HA and NA have been further purified by removal of other viral components). In one embodiment, the recombinant virus is grown in eggs and a split inactivated vaccine is prepared therefrom. In one embodiment, the dose in a vaccine contains about 10 to about 20, e.g., 15, μg of HA per strain (for example for a total HA concentration of 45 μg for trivalent and 60 μg for quadrivalent) and is administered as a single dose to those aged >9 years. Younger children (between 6 months and 8 years of age) may need two doses administered 4 weeks apart, if they have not been vaccinated in previous influenza seasons. The standard dose is typically delivered as an intramuscular (i.m.) injection (although intradermal [i.d.] formulations and intranasal formulations are also available).

[0078] In one embodiment, whole-virus vaccines are prepared from harvested allantoic fluid, chemically inactivated, e.g., with formalin or β -propiolactone, and subsequently concentrated and purified to remove nonviral protein contaminants. In one embodiment, the split-virus vaccine includes treatment with detergent to dissociate the viral lipid envelope, exposing all viral proteins and subviral elements. In one embodiment, for subunit vaccines, the HA (and NA) protein is further enriched through additional purification steps. Because the split-virus and subunit vaccines had comparable immunogenicity in primed populations but reduced reactogenicity compared to the whole-virus vaccine preparations, most contemporary vaccines since the 1970s have been split-virus or subunit formulations.

Exemplary NA Modifications

[0079] The present disclosure thus relates to influenza modification relative to parental NA that stabilize the neuraminidase (NA) tetramer, e.g., of human influenza viruses. Those NA modification(s) may also increase the vaccine virus yield.

[0080] Therefore, the disclosure provides isolated recombinant, e.g., reassortant, influenza viruses with selected amino acid residues, insertions, deletions, or any combination thereof, in the stalk region in NA. In one embodiment, the NA is selected to encode a cysteine at residue 48. In one embodiment, the NA is selected to encode a cysteine at position 50. In one embodiment, the NA is selected to encode a cysteine at positions 48 and 50.

[0081] In one embodiment, the NA is selected to have a deletion in one or more of residues 1 to 10 after the last residue in the transmembrane domain, which deletion stabilizes the NA tetramer. In one embodiment, the NA is selected to have a deletion in one or more of residues 10 to 20 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have a deletion in one or more of residues 20 to 30 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have a deletion in one or more of residues 30 to 40

after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have a deletion in one or more of residues 40 to 50 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the deletion includes a deletion of 1, 2, 3, 4 or 5 residues in the stalk region. In one embodiment, the deletion includes a deletion of 6, 7, 8, or 9 residues in the stalk region. In one embodiment, the deletion includes a deletion of 10, 11, 12, 13, 14 or 15 residues in the stalk region. In one embodiment, the deletion includes a deletion of 16, 17, 18, or 19 residues in the stalk region. In one embodiment, the deletion includes a deletion of 20, 21, 22, 23, 24 or 25 residues in the stalk region. In one embodiment, the deletion includes a deletion of 26, 27, 28, or 29 residues in the stalk region.

[0082] In one embodiment, the NA is selected to have an insertion of one or more amino acid residues in residues 1 to 10 after the last residue in the transmembrane domain, which insertion stabilizes the NA tetramer. In one embodiment, the NA is selected to have an insertion of one or more amino acid residues in residues 10 to 20 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have an insertion of one or more amino acid residues 20 to 30 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have an insertion of one or more amino acid residues 30 to 40 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have an insertion of one or more amino acid residues 40 to 50 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the insertion includes an insertion of 1, 2, 3, 4 or 5 residues in the stalk region. In one embodiment, the insertion includes an insertion of 6, 7, 8, or 9 residues in the stalk region. In one embodiment, the insertion includes an insertion of 10, 11, 12, 13, 14 or 15 residues in the stalk region. In one embodiment, the insertion includes an insertion of 16, 17, 18, or 19 residues in the stalk region. In one embodiment, the insertion includes an insertion of 20, 21, 22, 23, 24 or 25 residues in the stalk region. In one embodiment, the insertion includes an insertion of 26, 27, 28, or 29 residues in the stalk region.

[0083] In one embodiment, the NA is selected to have a substitution of one or more amino acid residues in residues 1 to 10 after the last residue in the transmembrane domain, which substitution stabilizes the NA tetramer. In one embodiment, the NA is selected to have a substitution of one or more amino acid residues in residues 10 to 20 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have a substitution of one or more amino acid residues 20 to 30 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have a substitution of one or more amino acid residues 30 to 40 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have a substitution of one or more amino acid residues 40 to 50 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the substitution to a cysteine is at a residue that faces towards the stalk region of a NA in another NA molecule, e.g., in a dimer, trimer or tetramer.

[0084] In one embodiment, the NA is selected to have one or more cysteines at one or more of residues 1 to 10 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have one or more cysteines at one or more of residues 10 to 20 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have one or more cysteines at one or more of residues 20 to 30 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have one or more cysteines at one or more of residues 30 to 40 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have one or more cysteines at one or more of residues 40 to 50 after the last residue in the transmembrane domain that stabilizes the NA tetramer. For example, a virus with a NA having a 7 amino acid cytoplasmic tail, a 26 amino acid transmembrane domain and a Cys at residue 48, is a virus that has a cysteine at a residue that is between residues 10 to 20 after the last residue in the transmembrane domain (the numbering for NA is based on N1). In one embodiment, the disclosure provides an isolated recombinant reassortant influenza virus having six “internal” viral segments from a vaccine influenza virus, e.g., PR8UW, a NA viral segment with one or more of the specified modifications, and a HA viral segment, e.g., any of H1-H18, e.g., from a circulating influenza virus. Also provided are compositions comprising the recombinant influenza virus, pharmaceutical compositions such as vaccines.

[0085] Thus, for vaccine viruses that are to be grown or passaged in cells, e.g., in eggs, replacement of the residue at any one of residues from 1 to 60 after the last residue in the transmembrane domain, an insertion of one or more residues from 1 to 60 after the last residue in the transmembrane domain or a deletion of one or more residues from 1 to 60 after the last residue in the transmembrane domain, or any combination thereof, in

NA, e.g., by mutation, or selection of a NA viral segment for a NA with a particular amino acid, e.g., cysteine, at any one of residues from 1 to 60 after the last residue in the transmembrane domain, an insertion of one or more residues from 1 to 60 after the last residue in the transmembrane domain or a deletion of one or more residues from 1 to 60 after the last residue in the transmembrane domain, or any combination thereof, in NA, wherein the numbering is based on N1, may result in stabilization of NA and/or higher viral titers. [0086] In one embodiment, the disclosure provides an isolated recombinant influenza virus comprising PA, PB1, PB2, NP, NS, M, and HA viral segments and a NA viral segment that encodes an NA selected to encode a particular amino acid, e.g., cysteine, at any one of residues from 1 to 60 after the last residue in the transmembrane domain, an insertion of one or more residues from 1 to 60 after the last residue in the transmembrane domain or a deletion of one or more residues from 1 to 60 after the last residue in the transmembrane domain. wherein the numbering is based on N1, wherein the recombinant influenza virus may have enhanced replication in avian eggs or a NA tetramer with enhanced stability, e.g., during vaccine production.

[0087] In one embodiment, the disclosure provides an isolated recombinant influenza virus comprising PA, PB1, PB2, NP, NS, M, and HA viral segments and a NA viral segment that encodes an NA with a replacement (substitution) of a residue at any one of residues from 1 to 60 after the last residue in the transmembrane domain, an insertion of one or more residues from 1 to 60 after the last residue in the transmembrane domain or a deletion of one or more residues from 1 to 60 after the last residue in the transmembrane domain, or any combination thereof, in NA, e.g., by mutation, wherein the numbering is based on N1, wherein the recombinant influenza virus may have enhanced replication in avian eggs or a NA tetramer with enhanced stability, e.g., during vaccine production.

[0088] In one embodiment, the isolated recombinant influenza virus is a reassortant. In one embodiment, the NA viral segment encodes a NA that has at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 99% amino acid sequence identity to any one of SEQ ID Nos. 1-9, 15-18, 37, or 44-50, or a polypeptide encoded by any one of SEQ ID Nos. 19-29, or having at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 99% amino acid sequence identity to the stalk region in any one of SEQ ID Nos. 1-9, 15-18, 37, or 44-50, or encoded by one of SEQ ID Nos. 19-29. In one embodiment, the NA viral segment encodes a N2, N3, N7, or N9 and the positions in N3, N7, or N9 with the specified modification(s). In one embodiment, the NA viral segment encodes a N1, N4, N5, N6, N8, N10 or N11 with the specified modification(s). In one embodiment, the PA, PB1, PB2, NP, M, and NS viral segments have at least 85% nucleic acid sequence identity to SEQ ID Nos. 30 to 35 or 38 to 43 or encode a polypeptide having at least 80%, 85%, 90%, 95%, or 99 amino acid sequence identity to a polypeptide encoded by SEQ ID Nos. 30-35 or 38 to 43. In one embodiment, the virus is an influenza B virus.

[0089] Also provided is a method to prepare influenza virus having a stabilized NA tetramer. The method includes contacting a cell with: a vector for vRNA production comprising a promoter operably linked to an influenza virus PA DNA linked to a transcription termination sequence, a vector for vRNA production comprising a promoter operably linked to an influenza virus PB1 DNA linked to a transcription termination sequence, a vector for vRNA production comprising a promoter operably linked to an influenza virus PB2 DNA linked to a transcription termination sequence, a vector for vRNA production comprising a promoter operably linked to an influenza virus HA DNA linked to a transcription termination sequence, a vector for vRNA production comprising a promoter operably linked to an influenza virus NP DNA linked to a transcription termination sequence, a vector for vRNA production comprising a promoter operably linked to an influenza virus NA DNA linked to a transcription termination sequence, a vector for vRNA production comprising a promoter operably linked to an influenza virus M DNA linked to a transcription termination sequence, and a vector for vRNA production comprising a promoter operably linked to an influenza virus NS DNA linked to a transcription termination sequence, wherein the PB1, PB2, PA, NP, NS, and M DNAs in the vectors for vRNA production are from one or more influenza vaccine virus isolates, wherein the NA DNA in the vector for vRNA production has a modification in the stalk region as described herein, wherein the numbering for NA residues is that for N1; and a vector for mRNA production comprising a promoter operably linked to a DNA segment encoding influenza virus PA, a vector for mRNA production comprising a promoter operably linked to a DNA segment encoding influenza virus PB1, a vector for mRNA production comprising a promoter operably linked to a DNA segment encoding influenza virus PB2, and a vector for mRNA production comprising a promoter operably linked to a DNA segment encoding influenza virus NP, and optionally comprising one or more of: a vector for mRNA production comprising a promoter operably linked to a DNA segment encoding influenza virus HA, a vector for mRNA production comprising a

promoter operably linked to a DNA segment encoding influenza virus NA, a vector for mRNA production comprising a promoter operably linked to a DNA segment encoding influenza virus M1, a vector for mRNA production comprising a promoter operably linked to a DNA segment encoding influenza virus M2, a vector for mRNA production comprising a promoter operably linked to a DNA segment encoding influenza virus NS1, or a vector for mRNA production comprising a promoter operably linked to a DNA segment encoding influenza virus NS2; in an amount effective to yield infectious influenza virus. In one embodiment, the NA viral segment encodes a NA that has at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 99% amino acid sequence identity to any one of SEQ ID Nos. 1-9, 15-18, 37, or 44-50, or having at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 99% amino acid sequence identity to the stalk region in any one of SEQ ID Nos. 1-9, 15-18, 37, or 44-50. In one embodiment, the NA is N1, N4, N5, N6, N8, N10 or N11.

[0090] In one embodiment, the HA is H1, H3, H5, H7, or H9. In one embodiment, the virus is an influenza A virus. In one embodiment, PA, PB1, PB2, NP, M, and NS viral segments have at least 85%, 85%, 90%, 95%, or 99% nucleic acid sequence identity to SEQ ID Nos. 30 to 35 or 38 to 43 or encode a polypeptide having at least 80%, 85%, 90%, 95%, or 99% amino acid sequence identity to a polypeptide encoded by SEQ ID Nos. 30-35 or 38 to 43. In one embodiment, HA is H2, H4, H5, H6, H8, or any of H10-H18.

[0091] In one embodiment, the virus is an influenza B virus.

[0092] Further provided is a method of immunizing an avian or a mammal with a composition having an effective amount of the virus described herein. In one embodiment, the composition comprises at least one other different influenza virus. In one embodiment, the mammal is a human. In one embodiment, the composition is administered intranasally or via injection.

[0093] In one embodiment, the disclosure provides isolated influenza type A virus with a characteristic residue(s), insertion and/or deletion, or a combination thereof, in NA described herein. In one embodiment, the isolated influenza type A virus with a characteristic residue(s), insertion and/or deletion, or a combination thereof, has a NA with at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 99% amino acid sequence identity to any one of SEQ ID Nos. 1-9, 15-18, 37, or 44-50, or a polypeptide encoded by any one of SEQ ID Nos. 19-29, or having at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 99% amino acid sequence identity to the stalk region in any one of SEQ ID Nos. 1-9, 15-18, 37, or 44-50 or encoded by one of SEQ ID Nos. 19-29.

[0094] In one embodiment, the isolated influenza type A virus of the invention with a characteristic residue(s) and/or deletion, or a combination thereof, has an HA from any one of subtypes 1-18 of HA. In one embodiment the characteristic residue is a conservative substitution. Conservative amino acid substitutions refer to the interchangeability of residues having similar side chains. For example, a group of amino acids having aliphatic side chains is glycine, alanine, valine, leucine, and isoleucine; a group of amino acids having aliphatic-hydroxyl side chains is serine and threonine; a group of amino acids having amide-containing side chains is asparagine and glutamine; a group of amino acids having aromatic side chains is phenylalanine, tyrosine and tryptophan; a group of amino acids having basic side chains is lysine, arginine and histidine; and a group of amino acids having sulfur-containing side chain is cysteine and methionine. In one embodiment, conservative amino acid substitution groups are: threonine-valine-leucine-isoleucine-alanine; phenylalanine-tyrosine; lysine-arginine; alanine-valine; glutamic-aspartic; and asparagine-glutamine. In one embodiment, the characteristic residue is a non-conservative substitution.

[0095] In one embodiment, a mutation is introduced into a NA viral segment of an influenza virus isolate, e.g., via recombinant DNA techniques including site-specific mutagenesis, or replacing a portion of the NA coding sequence with a portion that includes the characteristic residue(s), insertion or deletion. In one embodiment, a NA viral segment with a characteristic residue, insertion and/or deletion described herein is combined with a HA segment, and internal viral segments of an influenza vaccine virus.

[0096] The disclosure provides a plurality of influenza virus vectors of the invention, e.g., those useful to prepare reassortant viruses including 6:1:1 reassortants, 6:2 reassortants and 7:1 reassortants. A 6:1:1 reassortant is an influenza virus with 6 internal viral segments from a vaccine virus, a HA viral segment that is from a different (second) viral isolate than the vaccine virus, and a NA viral segment with a characteristic residue(s), insertion, and/or deletion, or a combination thereof, as described herein, which is from a different viral source than the HA segment and the vaccine virus; a 6:2 reassortant is an influenza virus with 6 internal viral segments from a vaccine virus, and a NA viral segment having a characteristic residue(s), insertion and/or deletion, or a combination thereof, which segment is from the same source as the HA segment, and a HA viral segment from a different viral isolate than the vaccine virus; and a 7:1 reassortant, in one

embodiment, is an influenza virus with 6 internal viral segments and a HA segment from a vaccine virus, and a NA segment that is modified to include the characteristic residue(s) and/or deletion, or a combination thereof, which NA segment is from a different viral source than the vaccine virus.

[0097] In one embodiment of the invention, the plurality includes vectors for vRNA production selected from a vector comprising a promoter operably linked to an influenza virus PA DNA linked to a transcription termination sequence, a vector comprising a promoter operably linked to an influenza virus PB1 DNA linked to a transcription termination sequence, a vector comprising a promoter operably linked to an influenza virus PB2 DNA linked to a transcription termination sequence, a vector comprising a promoter operably linked to an influenza virus HA DNA linked to a transcription termination sequence, a vector comprising a promoter operably linked to an influenza virus NP DNA linked to a transcription termination sequence, a vector comprising a promoter operably linked to an influenza virus NA DNA linked to a transcription termination sequence, a vector comprising a promoter operably linked to an influenza virus M DNA linked to a transcription termination sequence, and a vector comprising a promoter operably linked to an influenza virus NS DNA linked to a transcription termination sequence. In one embodiment, the DNAs for vRNA production of PB1, PB2, PA, NP, M, and NS, have sequences from an influenza virus that replicates to high titers in cultured mammalian cells such as MDCK cells, Vero cells or PER.C6® cells or embryonated eggs, and/or from a vaccine virus, e.g., one that does not cause significant disease in humans. The DNA for vRNA production of NA may be from any NA, e.g., any of N1-N11, and the DNA for vRNA production of HA may be from any HA, e.g., H1-H18. In one embodiment, the DNAs for vRNA production may be for an influenza B or C virus. For example, the DNAs for vRNA production include influenza B virus PA, PB1, PB2, NP, NS, and M or influenza B virus PA, PB1, PB2, NP, NS, M, and NA, wherein the vRNA for NA has a NA with a characteristic residue, insertion and/or deletion as described herein. The DNAs for vRNA production of NA and HA may be from different strains or isolates (6:1:1 reassortants) or from the same strain or isolate (6:2 reassortants), or the NA or HA may be from the same strain or isolate as that for the internal genes (7:1 reassortant). The plurality also includes vectors for mRNA production selected from a vector encoding influenza virus PA, a vector encoding influenza virus PB1, a vector encoding influenza virus PB2, and a vector encoding influenza virus NP, and optionally one or more vectors encoding NP, NS, M, e.g., M1 and M2, HA or NA. The vectors encoding viral proteins may further include a transcription termination sequence.

[0098] Viruses that may provide the internal genes for reassortants within the scope of the invention include viruses that have high titers, e.g., titers of at least about $10^{5.5}$ PFU/mL, e.g., at least $10^{6.0}$ PFU/mL, $10^{6.5}$ PFU/mL or $10^{6.8}$ PFU/mL; high titers in embryonated eggs, e.g., titers of at least about $10^{7.5}$ EID.sub.50/mL, e.g., at least $10^{8.0}$ EID.sub.50/mL, $10^{8.5}$ EID.sub.50/mL or $10^{9.0}$ EID.sub.50/mL; high titers in MDCK cells, e.g., titers of at least about $10^{6.5}$ PFU/mL, e.g., at least $10^{6.8}$ PFU/mL, or high titers in two of more of those host cells.

[0099] Other reassortants with internal genes from other PR8 isolates or vaccine viruses may be employed in recombinant reassortant viruses.

[0100] In one embodiment, the DNAs for the internal genes for PB1, PB2, PA, NP, M, and NS encode proteins with substantially the same activity as a corresponding polypeptide encoded by one of SEQ ID NOs:30-35 or 38-43. As used herein, “substantially the same activity” includes an activity that is about 0.1%, 1%, 10%, 30%, 50%, 90%, e.g., up to 100% or more, or detectable protein level that is about 80%, 90% or more, the activity or protein level, respectively, of the corresponding full-length polypeptide. In one embodiment, the nucleic acid a sequence encoding a polypeptide which is substantially the same as, e.g., having at least 80%, e.g., 90%, 92%, 95%, 97% or 99%, including any integer between 80 and 99, contiguous amino acid sequence identity to, a polypeptide encoded by one of SEQ ID NOs:30-35 or 38-43. In one embodiment, the isolated and/or purified nucleic acid molecule comprises a nucleotide sequence which is substantially the same as, e.g., having at least 50%, e.g., 60%, 70%, 80% or 90%, including any integer between 50 and 100, or more contiguous nucleic acid sequence identity to one of SEQ ID NOs:30-35 or 38-43 and, in one embodiment, also encodes a polypeptide having at least 80%, e.g., 90%, 92%, 95%, 97% or 99%, including any integer between 80 and 99, contiguous amino acid sequence identity to a polypeptide encoded by one of SEQ ID NOs:30-35 or 38-43. In one embodiment, the influenza virus polypeptide has one or more, for instance, 2, 5, 10, 15, 20 or more, conservative amino acids substitutions, e.g., conservative substitutions of up to 10% or 20% of the residues, relative to a polypeptide encoded by one of SEQ ID NOs:30-35 or 38-43. Conservative amino acid substitutions refer to the interchangeability of residues having similar side chains. For example, a group of amino acids having aliphatic side chains is glycine, alanine, valine, leucine, and isoleucine; a group of amino acids having aliphatic-hydroxyl side chains is serine and

threonine; a group of amino acids having amide-containing side chains is asparagine and glutamine; a group of amino acids having aromatic side chains is phenylalanine, tyrosine and tryptophan; a group of amino acids having basic side chains is lysine, arginine and histidine; and a group of amino acids having sulfur-containing side chain is cysteine and methionine. In one embodiment, conservative amino acid substitution groups are: valine-leucine-isoleucine; phenylalanine-tyrosine; lysine-arginine; alanine-valine; glutamic-aspartic; and asparagine-glutamine. In one embodiment, the influenza virus polypeptide has one or more, for instance, 2, 3 or 4, nonconservative amino acid substitutions, relative to a polypeptide encoded by one of SEQ ID NOs:30-35.

[0101] In one embodiment, the nucleic acid is a sequence encoding a NA polypeptide which is substantially the same as, e.g., having at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, e.g., 90%, 92%, 95%, 97% or 99%, including any integer between 80 and 99, contiguous amino acid sequence identity to, one of Accession Nos. ACP41107.1 (N1), AIK26357.1 (N7), ALH21372.1 (N9), or BAK86313.1 (N2), the sequences of which are incorporated by reference herein, or at least the stalk region thereof. In one embodiment, the isolated and/or purified nucleic acid molecule encodes a polypeptide having at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, e.g., 90%, 92%, 95%, 97% or 99%, including any integer between 80 and 99, contiguous amino acid sequence identity to any one of Accession Nos. ACP41107.1 (N1), AIK26357.1 (N7), ALH21372.1 (N9), or BAK86313.1 (N2), the sequences of which are incorporated by reference herein, or at least the stalk region thereof. In one embodiment, the influenza virus polypeptide has one or more, for instance, 2, 5, 10, 15, 20 or more, conservative amino acids substitutions, e.g., conservative substitutions of up to 10% or 20% of the residues, relative to SEQ ID NOs:1-18, or one of Accession Nos. ACP41107.1 (N1) AIK26357.1 (N7), ALH21372.1 (N9), or BAK86313.1 (N2), or at least the stalk region thereof. Conservative amino acid substitutions refer to the interchangeability of residues having similar side chains. For example, a group of amino acids having aliphatic side chains is glycine, alanine, valine, leucine, and isoleucine; a group of amino acids having aliphatic-hydroxyl side chains is serine and threonine; a group of amino acids having amide-containing side chains is asparagine and glutamine; a group of amino acids having aromatic side chains is phenylalanine, tyrosine and tryptophan; a group of amino acids having basic side chains is lysine, arginine and histidine; and a group of amino acids having sulfur-containing side chain is cysteine and methionine. In one embodiment, conservative amino acid substitution groups are: valine-leucine-isoleucine; phenylalanine-tyrosine; lysine-arginine; alanine-valine; glutamic-aspartic; and asparagine-glutamine. In one embodiment, the influenza virus polypeptide has one or more, for instance, 2, 3 or 4, nonconservative amino acid substitutions, relative to a polypeptide having one of SEQ ID NOs:1-18, 3, or one of Accession Nos. ACP41107.1 (N1) AIK26357.1 (N7), ALH21372.1 (N9), or BAK86313.1 (N2), or at least the stalk region thereof.

[0102] The invention thus includes the use of isolated and purified vectors or plasmids, which express or encode influenza virus proteins, or express or encode influenza vRNA, both native and recombinant vRNA. The vectors comprise influenza cDNA, e.g., influenza A (e.g., any influenza A gene including any of the 18 HA or 11 NA subtypes), B or C DNA (see Fields *Virology* (Fields et al. (eds.), Lippincott, Williams and Wickens (2013), which is specifically incorporated by reference herein). Any suitable promoter or transcription termination sequence may be employed to express a protein or peptide, e.g., a viral protein or peptide, a protein or peptide of a nonviral pathogen, or a therapeutic protein or peptide.

[0103] A composition or plurality of vectors of the invention may also comprise a heterologous gene or open reading frame of interest, e.g., a foreign gene encoding an immunogenic peptide or protein useful as a vaccine or in gene replacement, for instance may encode an epitope useful in a cancer therapy or vaccine, or a peptide or polypeptide useful in gene therapy. When preparing virus, the vector or plasmid comprising the gene or cDNA of interest may substitute for a vector or plasmid for an influenza viral gene or may be in addition to vectors or plasmids for all influenza viral genes. Thus, another embodiment of the invention comprises a composition or plurality of vectors as described above in which one of the vectors is replaced with, or further comprises, 5' influenza virus sequences optionally including 5' influenza virus coding sequences or a portion thereof, linked to a desired nucleic acid sequence, e.g., a desired cDNA, linked to 3' influenza virus sequences optionally including 3' influenza virus coding sequences or a portion thereof. In one embodiment, the desired nucleic acid sequence such as a cDNA is in an antisense (antigenomic) orientation. The introduction of such a vector in conjunction with the other vectors described above to a host cell permissive for influenza virus replication results in recombinant virus comprising vRNA corresponding to the heterologous sequences of the vector.

[0104] The promoter in a vector for vRNA production may be a RNA polymerase I promoter, a RNA

polymerase II promoter, a RNA polymerase III promoter, a T7 promoter, or a T3 promoter, and optionally the vector comprises a transcription termination sequence such as a RNA polymerase I transcription termination sequence, a RNA polymerase II transcription termination sequence, a RNA polymerase III transcription termination sequence, or a ribozyme. Ribozymes within the scope of the invention include, but are not limited to, tetrahymena ribozymes, RNase P, hammerhead ribozymes, hairpin ribozymes, hepatitis ribozyme, as well as synthetic ribozymes. In one embodiment, the RNA polymerase I promoter is a human RNA polymerase I promoter.

[0105] The promoter or transcription termination sequence in a vRNA or virus protein expression vector may be the same or different relative to the promoter or any other vector. In one embodiment, the vector or plasmid which expresses influenza vRNA comprises a promoter suitable for expression in at least one particular host cell, e.g., avian or mammalian host cells such as canine, feline, equine, bovine, ovine, or primate cells including human cells, or for expression in more than one host.

[0106] In one embodiment, at least one vector for vRNA comprises a RNA polymerase II promoter linked to a ribozyme sequence linked to viral coding sequences linked to another ribozyme sequences, optionally linked to a RNA polymerase II transcription termination sequence. In one embodiment, at least 2, e.g., 3, 4, 5, 6, 7 or 8, vectors for vRNA production comprise a RNA polymerase II promoter, a first ribozyme sequence, which is 5' to a sequence corresponding to viral sequences including viral coding sequences, which is 5' to a second ribozyme sequence, which is 5' to a transcription termination sequence. Each RNA polymerase II promoter in each vRNA vector may be the same or different as the RNA polymerase II promoter in any other vRNA vector. Similarly, each ribozyme sequence in each vRNA vector may be the same or different as the ribozyme sequences in any other vRNA vector. In one embodiment, the ribozyme sequences in a single vector are not the same.

[0107] In one embodiment, at least one vector comprises sequences corresponding to those encoding PB1, PB2, PA, NP, M, or NS, or a portion thereof, having substantially the same activity as a corresponding polypeptide encoded by one of SEQ ID NOs:30-35, e.g., a sequence encoding a polypeptide with at least 80%, e.g., 85%, 90%, 92%, 95%, 98%, 99% or 100%, including any integer between 80 and 100, amino acid identity to a polypeptide encoded by one of SEQ ID NOs:30-35. Optionally, two vectors may be employed in place of the vector comprising a promoter operably linked to an influenza virus M cDNA linked to a transcription termination sequence, e.g., a vector comprising a promoter operably linked to an influenza virus M1 cDNA linked to a transcription termination sequence and a vector comprising a promoter operably linked to an influenza virus M2 cDNA linked to a transcription termination sequence.

[0108] A plurality of the vectors of the invention may be physically linked or each vector may be present on an individual plasmid or other, e.g., linear, nucleic acid delivery vehicle. In one embodiment, each vRNA production vector is on a separate plasmid. In one embodiment, each mRNA production vector is on a separate plasmid.

[0109] The invention also provides a method to prepare influenza virus. The method comprises contacting a cell with a plurality of the vectors of the invention, e.g., sequentially or simultaneously, in an amount effective to yield infectious influenza virus. The invention also includes isolating virus from a cell contacted with the plurality of vectors. Thus, the invention further provides isolated virus, as well as a host cell contacted with the plurality of vectors or virus of the invention. In another embodiment, the invention includes contacting the cell with one or more vectors, either vRNA or protein production vectors, prior to other vectors, either vRNA or protein production vectors. In one embodiment, the promoter for vRNA vectors employed in the method is a RNA polymerase I promoter, a RNA polymerase II promoter, a RNA polymerase III promoter, a T3 promoter or a T7 promoter. In one embodiment, the RNA polymerase I promoter is a human RNA polymerase I promoter. In one embodiment, each vRNA vector employed in the method is on a separate plasmid. In one embodiment, the vRNA vectors employed in the method are on one plasmid or on two or three different plasmids. In one embodiment, each mRNA vector employed in the method is on a separate plasmid. In one embodiment, the mRNA vectors for PA, PB1, PB2 and NP employed in the method are on one plasmid or on two or three different plasmids.

[0110] The methods of producing virus described herein, which do not require helper virus infection, are useful in viral mutagenesis studies, and in the production of vaccines (e.g., for AIDS, influenza, hepatitis B, hepatitis C, rhinovirus, filoviruses, malaria, herpes, and foot and mouth disease) and gene therapy vectors (e.g., for cancer, AIDS, adenosine deaminase, muscular dystrophy, ornithine transcarbamylase deficiency and central nervous system tumors). Thus, a virus for use in medical therapy (e.g., for a vaccine or gene therapy) is provided.

[0111] The invention also provides isolated viral polypeptides and using recombinant virus of the invention. The methods include administering to a host organism, e.g., a mammal, an effective amount of the influenza virus of the invention, e.g., an inactivated virus preparation, optionally in combination with an adjuvant and/or a carrier, e.g., in an amount effective to prevent or ameliorate infection of an animal such as a mammal by that virus or an antigenically closely related virus. In one embodiment, the virus is administered intramuscularly while in another embodiment, the virus is administered intranasally. In some dosing protocols, all doses may be administered intramuscularly or intranasally, while in others a combination of intramuscular and intranasal administration is employed. The vaccine may further contain other isolates of influenza virus including recombinant influenza virus, other pathogen(s), additional biological agents or microbial components, e.g., to form a multivalent vaccine. In one embodiment, intranasal vaccination, for instance containing with inactivated influenza virus, and a mucosal adjuvant may induce virus-specific IgA and neutralizing antibody in the nasopharynx as well as serum IgG.

[0112] The influenza virus of the invention may employed with other anti-virals, e.g., amantadine, rimantadine, and/or neuraminidase inhibitors, e.g., may be administered separately in conjunction with those anti-virals, for instance, administered before, during and/or after.

[0113] One example of an influenza A virus (A/Yokohama/2013/2003(H3N2)) neuraminidase protein sequence and its stalk region is provided below (stalk region is underlined).

TABLE-US-00001 (SEQ ID NO: 1) MNPNQKIITI GSVSLTISTI CFFMQIAILI TTVTLHFKQY
EFNSPPNNQV MLCEPTIER NITEIVYLTN TTIEKEICPK LAEYRNWSKP QCNITGFAPF
SKDNSIRLSA GGDIWVTREP YVSCDPDKCY QFALGQGTTL NNVHSNDIVH DRTPYRTLLM
NELGVPFHLG TKQVCIAWSS SSCHDGKAWL HVCVTGDDEN ATASFIYNGR
LADSIVSWSK KILRTQESEC VCINGTCTVV MTDGSASGKA DTKILFIEEG KIVHTSTLSG
SAQHVEECSC YPRYPGVRCV CRDNWKGSNR PIVDINIKDY SIVSSYVCSG LVGDTPRKND
SSSSSHCLDP NNEEGGHGVK GWAFFDDGNDV WMGRTISEKL RSGYETFKVI
EGWSNPNSKL QINRQVIVDR GNRSGYSGIF SVEGKSCINR CFYVELIRGR KQETEVLTWS
NSIVVFCGTS GTYGTGSWPD GADINLMPI

Another example of an influenza A virus (A/Yokohama/47/2002(H1N2)) neuraminidase sequence and its stalk region is shown below (stalk region is underlined).

TABLE-US-00002 (SEQ ID NO: 3) MNPNQKIITI GSVSLTIATI CFLMQIAILV
TTVTLHFKQY ECNSPPNNQV MLCEPTIER NITEIVYLTN TTIEKEICPK LAEYRNWSKP
QCNITGFAPF SKDNSIRLSA GGDIWVTREP YVSCDPDKCY QFALGQGTTL
NNGHSNDTVH DRTPYRTLLM NELGVPFHLG TKQVCIAWSS SSCHDGKAWL
HVCVTGDDGN ATASFIYNGR LVDSIGSWSK KILRTQESEC VCINGTCTVV MTDGSASGKA
DTKILFIEEG KIVHTSLLSG SAQHVEECSC YPRYPGVRCV CRDNWKGSNR PIVDINVKDY
SIVSSYVCSG LVGDTPRKND SSSSHCLDP NNEEGGHGVK GWAFFDDGNDV
WMGRTISEKL RSGYETFKVI EGWSKPNLSL QINRQVIVDR GNRSGYSGIF SVEGKSCINR
CFYVELIRGR NQETEVLTWS NSIVVFCGTS GTYGTGSWPD GADINLMPI

The amino acid sequence for Singapore0019 NA and its stalk region is as follows (stalk region is underlined):

TABLE-US-00003 (SEQ ID NO: 2) M N P N Q K I I T I G S V S L T I
S T I C F F M Q I A I L I T T V T L H F K Q Y E F N S
P P N N Q V M L C E P T I I E R N I T E I V Y L T N T
T I E K E I C P K P A E Y R N W S K P Q C G I T G F A
P F S K D N S I R L S A G G D I W V T R E P Y V S C D
P D K C Y Q F A L G Q G T T L N N V H S N N T V R D
R T P Y R T L L M N E L G V P F H L G T K Q V C I A
W S S S S C H D G K A W L H V C I T G D D K N A T A
S F I Y N G R L I D S V V S W S K D I L R T Q E S E C
V C I N G T C T V V M T D G N A T G K A D T K I L F I
E E G K I V H T S K L S G S A Q H V E E C S C Y P R Y
P G V R C V C R D N W K G S N R P I V D I N I K D H S
I V S S Y V C S G L V G D T P R K N D S S S S S H C L
N P N N E E G G H G V K G W A F D D G N D V W M G
R T I N E T S R L G Y E T F K V V E G W S N P K S K L
Q I N R Q V I V D R G D R S G Y S G I F S V E G K S C
I N R C F Y V E L I R G R K E E T E V L W T S N S I V

V F C G T S G T Y G T G S W P D G A D L N M H I.

The NA for A/Alaska/232/2015 has the following sequence (stalk region is underlined):

TABLE-US-00004 (SEQ ID NO: 44) mnpnqiiti gsvltisti cffmqiaili ttvtlhfky_efnspnnqv
mlceptiier niteivyltn tteikeicpk paeyrnwskp qcgitgfapf skdnsirlsa ggdiwvtrep yvscdpdkcy
qfalgggttl nnvhsnntvr **dr**tpyrllm nelgvpfhlg tkqvciawss sschdgkawl hvcitgddkn atasfiyngr
lvdsvvswsk dilrtqesec vcingtctvv mtd**gn**atgka dtkilfieeg kivhtsklsg saqhveecsc yprypgvrcv
crdnwkgshr pivdinikdh sivssyvcsg lvgdtpkrnd ssssshclnp nneeg**gh**gvk gwafddgndv wmgrtinets
rlgyetfkvv egwsnpkskl qinrvivdr gdrsgysgif svegkscinr cfyvelirgr keetevlwts nsivvcgts
gtygtgswpd gadnlmhi. NA nucleotide sequence for A/Singapore/GP1908/2015 (H1N1) pdm09
(SEQ ID NO: 36)

agtttaaatgaatccaaacaaaagataataaccattgggtcgatcagatgacaattggaatggctaacttaattacaaattgg
aaacataatctcaatatgggttagccactcaattcaaattggaaatcaaagccagattgaacatgcaatcaaagcgctacttatt
gaaaacaacacttgggtaaatcagacatatgttaacatcagcaacaccaactttgctgctggacagtcagtggttccgtgaaattag
cgggcaattcctctctctgcctgttagtggtgggtatatacagtaaagacaacagtgtaagaatcgggtccaaggggagtggtt
tgtcataaggaaccattcatatcatgctctccttggaaatgcagaaccttctcttgactcaaggggccttgctaaatgacaacat
tccaatggaaccattaaagacaggagcccatatcgaaacctaatgagctgtcctattggtgaagttcctctccatacaactcaagat
ttgagtcagtcgcttggtcagcaagtgctgtcatgatggcatcaattggctaacaattggaatttctggcccagacagtggggcagt
ggctgtgttaaagtacaatggcataataacagacactatcaagagttggaggaacaatatattgagaacacaagagtctgaatgtgca
tgtgtaaattggttctgtttaccataatgaccgatggaccaagtgtgagcaggcctatacaaaaattcagaatagaaaaggaa
agataatcaaatcagtcgaaatgaaagcccctaattatcactatgaggaatgctcctgttacctgatttagtgaaatcacatgtgt
gtgcagggataactggcatggctcgaaatcgaccgtgggtgtcttcaaccagaatctggaatatcagatgggatacatatgcagtggg
gttttcggagacaatccacgccctaataagacaggcagttgtggtccagtatcgtctaattggagcaaatggagtaaaaggatttt
cattcaaatacggcaatggtgtttggatagggagaactaaaagcattagttaagaaaagggtttgagatgattgggatccgaatgg
atggactgggactgacaataaattctcaataaagcaagatcgttaggaataaatgagtggtcagggtatagcgggagttttgtcag
catccagaactaacagggtcgattgtataagacctgtctctgggtgaactaataagagggcgacccgaagagaacacaatctgga
ctagcgggagcagcatatcctttgtggtgtaaacagtgacactgtgggttggtcttgccagacgggtgctgagtgccattaccat
tgacaagtaatttgtcaaaaaact Amino acid sequence for NA of A/Singapore/GP1908/2015 (H1N1)
pdm09 (stalk is underlined) (SEQ ID NO: 37)

MNPNQKIITIGSISMTIGMANLILQIGNIISIWVSH**SIQIGNQSQIETCNQSVITYENNTWVN**
QTYVNISNTNFAAGQSVVSVKLAGNSSLC P V S G W A I Y S K D N S V R I G S K G D V F V I R E P F I S C
S P L E C R T F F L T Q G A L L N D K H S N G T I K D R S P Y R T L M S C P I G E V P S P Y N S R F E S V A W S A S A C H
D G I N W L T I G I S G P D S G A V A V L K Y N G I I T D T I K S W R N N I L R T Q E S E C A C V N G S C F T I M T D G P
S D G Q A S Y K I F R I E K G K I I K S V E M K A P N Y H Y E E C S C Y P D S S E I T C V C R D N W H G S N R P W V S F
N Q N L E Y Q M G Y I C S G V F G D N P R P N D K T G S C G P V S S N G A N G V K G F S F K Y G N G V W I G R T K S
I S S R K G F E M I W D P N G W T G T D N K F S I K Q D I V G I N E W S G Y S G S F V Q H P E L T G L D C I R P C F W V
E L I R G R P E E N T I W T S G S S I S F C G V N S D T V G W S W P D G A E L P F T I D K

[0114] In some cases, in one or more modifications can also be introduced into HA, PA, PB1, PB2, NP, M1, M2, NS2, PB1-F2, PA-X, and/or NS1 proteins (and nucleic acids encoding such proteins).

[0115] Besides enhanced stability during vaccine production, enhanced growth of the virus when passaged through embryonated chicken eggs or cultured cells may be observed when the modified NA proteins are expressed and such expression may result in significantly higher viral titers. Thus, the invention provides a method for making influenza viruses with enhanced replication in cell culture or embryonated eggs. The method includes providing cells suitable for influenza vaccine production; infecting the cells with viruses having modified neuraminidase; and isolating virus strains with enhanced growth relative to the one or more unmodified viral isolates. In some cases, a method for making influenza viruses with enhanced replication in cell culture can involve, serially culturing one or more influenza virus isolates in embryonated chicken eggs; and isolating serially cultured virus with enhanced growth relative to the one or more isolates prior to serial culture. In some cases, the viruses can be grown or passaged within cells in culture, e.g., MDCK or Vero cells.

[0116] As discussed herein, the modified neuraminidases can be expressed in a variety of influenza strains. For example, A/Puerto Rico/8/34 (H1N1), “PR8,” virus often serves as the genetic backbone for generation of inactivated influenza vaccines.

[0117] In one embodiment, vectors for vRNA production can include a vector comprising a promoter operably linked to a modified NA DNA linked to a transcription termination sequence, a vector comprising a promoter operably linked to an influenza virus PB1 DNA linked to a transcription termination sequence, a

vector comprising a promoter operably linked to an influenza virus PB2 DNA linked to a transcription termination sequence, a vector comprising a promoter operably linked to an influenza virus HA DNA linked to a transcription termination sequence, a vector comprising a promoter operably linked to an influenza virus NP DNA linked to a transcription termination sequence, a vector comprising a promoter operably linked to an influenza virus NA DNA linked to a transcription termination sequence, a vector comprising a promoter operably linked to an influenza virus M DNA linked to a transcription termination sequence, and a vector comprising a promoter operably linked to an influenza virus NS DNA linked to a transcription termination sequence. In one embodiment, the DNAs for vRNA production of PB1, PB2, PA, NP, M, and NS, have sequences from an influenza virus that replicates to high titers in cultured mammalian cells such as MDCK cells, Vero cells or PER.C6® cells or embryonated eggs, and/or from a vaccine virus, e.g., one that does not cause significant disease in humans. The DNA for vRNA production of NA may be from any NA, e.g., any of N1-N11, and the DNA for vRNA production of HA may be from any HA, e.g., H1-H18. In one embodiment, the DNAs for vRNA production may be for an influenza B or C virus. The DNAs for vRNA production of NA and HA may be from different strains or isolates (6:1:1 reassortants) or from the same strain or isolate (6:2 reassortants), or the NA may be from the same strain or isolate as that for the internal genes (7:1 reassortant). Vectors for mRNA production can include a vector encoding a modified NA, a vector encoding influenza virus PA, a vector encoding influenza virus PB1, a vector encoding influenza virus PB2, and a vector encoding influenza virus NP, and optionally one or more vectors encoding NP, NS, M, e.g., M1 and M2, HA or NA. The vectors encoding viral proteins may further include a transcription termination sequence.

[0118] Other reassortants with internal genes from other PR8 isolates or vaccine viruses may be employed in recombinant reassortant viruses of the invention. In particular, 5:1:2 reassortants having UW-PR8 PB1, PB2, PA, NP, and M (“5”) and PR8(Cam) NS (“1”); 6:1:1 reassortants having UW-PR8 (modified) NA, PB1, PB2, PA, NP, and M (“6”) and PR8(Cam) NS (“1”); and 7:1 reassortants having UW-PR8 PB1, PB2, PA, NP, M, (modified) NA, and NS (“7”) may be employed.

[0119] The neuraminidases that can be modified can have any sequence including but not limited to the sequences described herein. However, in some cases, the modified neuraminidases can have substantially the same activity as a corresponding polypeptide described by sequence herein. As used herein, “substantially the same activity” includes an activity that is about 0.1%, 1%, 10%, 30%, 50%, 90%, e.g., up to 100% or more activity, or a detectable protein level that is about 80%, 90% or more protein level, of the corresponding protein described herein. In one embodiment, the nucleic acid encodes a polypeptide which is substantially the same as, e.g., having at least 80%, e.g., 90%, 92%, 95%, 97%, 98%, or 99%, including any integer between 80 and 99, contiguous amino acid sequence identity to a polypeptide encoded by one of sequences described herein. In one embodiment, the isolated and/or purified nucleic acid molecule comprises a nucleotide sequence which is substantially the same as, e.g., having at least 50%, e.g., 60%, 70%, 80% or 90%, including any integer between 50 and 100, or more contiguous nucleic acid sequence identity to one of the nucleic acid sequences described herein. In one embodiment, a nucleic acid also encodes a polypeptide having at least 80%, e.g., 90%, 92%, 95%, 97%, 98%, or 99%, including any integer between 80 and 99, contiguous amino acid sequence identity to a polypeptide described herein.

[0120] Exemplary viral sequences for a master vaccine strain (PR8UW) are as follows:

TABLE-US-00005 PA (SEQ ID NO: 30) AGCGAAAGCA GGTACTGATC CAAAATGGAA
GATTTTGTGC GACAATGCTT CAATCCGATG ATTGTCGAGC TTGCGGAAAA
AACAATGAAA GAGTATGGGG AGGACCTGAA AATCGAAACA AACAAATTTC
CAGCAATATG CACTCACTTG GAAGTATGCT TCATGTATTC AGATTTTCAC
TTCATCAATG AGCAAGGCGA GTCAATAATC GTAGAACTTG GTGATCCAAA
TGCACCTTTTG AAGCACAGAT TTGAAATAAT CGAGGGAAGA GATCGCACAA
TGGCCTGGAC AGTAGTAAAC AGTATTTGCA AACTACAGG GGCTGAGAAA
CCAAAGTTTC TACCAGATTT GTATGATTAC AAGGAGAATA GATTCATCGA
AATTGGAGTA ACAAGGAGAG AAGTTCACAT ATACTATCTG GAAAAGGCCA
ATAAAATTAA ATCTGAGAAA ACACACATCC ACATTTTCTC GTTCACTGGG
GAAGAAATGG CCACAAAGGC AGACTACACT CTCGATGAAG AAAGCAGGGC
TAGGATCAAA ACCAGACTAT TCACCATAAG ACAAGAAATG GCCAGCAGAG
GCCTCTGGGA TTCCTTTTCGT CAGTCCGAGA GAGGAGAAGA GACAATTGAA
GAAAGGTTTG AAATCACAGG AACAATGCGC AAGCTTGCCG ACCAAAGTCT
CCCGCCGAAC TTCTCCAGCC TTGAAAATTT TAGAGCCTAT GTGGATGGAT
TCGAACCGAA CGGCTACATT GAGGGCAAGC TGTCTCAAAT GTCCAAAGAA

GTAAATGCTA GAATTGAACC TTTTTTGAAA ACAACACCAC GACCACTTAG
ACTTCCGAAT GGGCCTCCCT GTTCTCAGCG GTCCAAATTC CTGCTGATGG
ATGCCTTAAA ATTAAGCATT GAGGACCCAA GTCATGAAGG AGAGGGAATA
CCGCTATATG ATGCAATCAA ATGCATGAGACATTCTTTG GATGGAAGGA
ACCCAATGTT GTTAAACCAC ACGAAAAGGG AATAAATCCA AATTATCTTC
TGTCATGGAA GCAAGTACTG GCAGAACTGC AGGACATTGA GAATGAGGAG
AAAATTCCAA AGACTAAAAA TATGAAGAAA ACAAGTCAGC TAAAGTGGGC
ACTTGGTGAG AACATGGCAC CAGAAAAGGT AGACTTTGAC GACTGTAAAG
ATGTAGGTGA TTTGAAGCAA TATGATAGTG ATGAACCAGA ATTGAGGTGC
CTTGCAAGTT GGATTCAGAA TGAGTTTAAAC AAGGCATGCG AACTGACAGA
TTCAAGCTGG ATAGAGCTCG ATGAGATTGG AGAAGATGTG GCTCCAATTG
AACACATTGC AAGCATGAGA AGGAATTATT TCACATCAGA GGTGTCTCAC
TGCAGAGCCA CAGAATACAT AATGAAGGGA GTGTACATCA ATACTGCCTT
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Exemplary Embodiments

[0121] In one embodiment, an isolated recombinant influenza virus comprising a neuraminidase (NA) viral segment encoding a NA monomer that forms virions having stabilized NA tetramers is provided. In one embodiment, the recombinant influenza virus has a modified NA stalk that results in stabilized tetramers relative to an influenza virus having a NA with an unmodified NA stalk. In one embodiment, the modified NA stalk has a deletion. In one embodiment, the modified NA stalk has an insertion. In one embodiment, the modified NA stalk has at least one amino acid substitution relative to the unmodified stalk. In one embodiment, the modified stalk has two or more of: a deletion, an insertion, or at least one amino acid substitution. In one embodiment, the at least one substitution in the modified NA stalk is a cysteine substitution. In one embodiment, the modified NA stalk has at least two substitutions. In one embodiment, the NA has a cysteine at position 48 relative to the numbering of N1. In one embodiment, the NA has a cysteine at position 50 relative to the numbering of N1. In one embodiment, the NA has a cysteine at position 48 and position 50 relative to the numbering of N1. In one embodiment, the NA stalk is modified within residues 1 to 10 from the C-terminus of the transmembrane domain. In one embodiment, the NA stalk is modified within residues 10 to 20 from the C-terminus of the transmembrane domain. In one embodiment, the NA stalk is modified within residues 20 to 30 from the C-terminus of the transmembrane domain. In one

embodiment, the NA stalk is modified within residues 30 to 50 from the C-terminus of the transmembrane domain.

[0122] For example, the recombinant influenza virus may have a N1 NA, and that N2 may have one or two cysteines in the stalk region, e.g., spaced apart with at least one residue in between, e.g., the stalk may have the following sequence:

TABLE-US-00006 (SEQ ID NO: 51)

SIQIGNQSQIETCNQSVITYENNTWVNQTYVNISNTNFAAGQSVVSVKL AGNSS, which could be modified to, for example, (SEQ ID NO: 52)

IQIGNQSQIECCNQSVITYENNTWVNQTYVNISNTNFAAGQSVVSVKLA GNSS, (SEQ ID NO: 53) IQIGNQSQIETCCQSVITYENNTWVNQTYVNISNTNFAAGQSVVSVKLA GNSS, (SEQ ID NO: 54) IQIGNQSQIECCCQSVITYENNTWVNQTYVNISNTNFAAGQSVVSVKLA GNSS, (SEQ ID NO: 55) IQIGNQSQICTCNQSVITYENNTWVNQTYVNISNTNFAAGQSVVSVKLA GNSS, or (SEQ ID NO: 56) IQIGNQSQIETCNCSVITYENNTWVNQTYVNISNTNFAAGQSVVSVKLA GNSS.

[0123] In one embodiment, a vaccine comprising an effective amount of the recombinant influenza virus or a portion thereof is provided. In one embodiment, the vaccine is a whole virus vaccine. In one embodiment, the vaccine is a split virus vaccine. In one embodiment, the vaccine is a subunit vaccine. In one embodiment, the vaccine further comprises an adjuvant. In one embodiment, the vaccine further comprises a pharmaceutically acceptable carrier. In one embodiment, the carrier is suitable for intranasal or intramuscular administration. In one embodiment, the vaccine further comprises at least one other influenza virus isolate. In one embodiment, the vaccine further comprises at least one other microbe or microbial antigen, e.g., a non-influenza virus, a bacterial or fungal antigen.

[0124] In one embodiment, a method of preparing influenza virus having stabilized NA tetramers is provided. The method includes contacting a cell with one or more vectors comprising nucleic acid for an influenza virus NA segment encoding a NA monomer that forms virions having stabilized NA tetramers, nucleic acid for an influenza virus PA segment, nucleic acid for an influenza virus PB1 segment, nucleic acid for an influenza virus PB2 segment, nucleic acid for an influenza virus NP segment, nucleic acid for an influenza virus NS segment, nucleic acid for an influenza virus M segment, and nucleic acid for an influenza virus HA segment, in an amount effective to produce influenza virus having stabilized NA tetramers. In one embodiment, the NA is N1, N2, N3 or N5. In one embodiment, the cell is a mammalian cell. In one embodiment, the cell is a 293T, PER.C6®, MDCK, MvLu1, CHO or Vero cell, or a cell in an avian egg.

[0125] In one embodiment, a method of making an influenza vaccine is provided comprising: providing the recombinant virus; and combining the virus with an adjuvant or treating the virus with an agent that inactivates the virus. In one embodiment, the method includes aliquoting a dose of the virus into a receptacle. In one embodiment, the adjuvant comprises immunostimulatory DNA sequences, bacterium-derived components, aluminum salt (alum) or squalene oil-in-water emulsion systems such as MF59 and AS03. In one embodiment, wherein the agent chemically inactivates the virus. In one embodiment, the agent comprises formalin or beta-propiolactone. In one embodiment, the agent comprises a detergent. In one embodiment, the detergent is a non-ionic detergent. In one embodiment, the detergent is a cationic detergent. In one embodiment, the detergent is an anionic detergent. In one embodiment, the detergent comprises CTAB, ammonium deoxycholate, Triton, SDS, Neodol 23-6, or sodium desoxycholate. In one embodiment, the agent comprises ether. In one embodiment, the method further comprises separating HA and NA from other viral components.

[0126] In one embodiment, a method of preparing influenza virus is provided comprising: contacting cells with the recombinant virus in an amount effective to yield progeny virus. In one embodiment, the virus is contacted with an avian egg. In one embodiment, the cells are mammalian cells. In one embodiment, the HA of the virus is H1, H3, H5 or H7.

[0127] Further provided is a method of preparing stabilized NA tetramers. The method includes contacting a cell with one or more vectors comprising nucleic acid for an influenza virus NA segment encoding a NA monomer that forms virions having stabilized NA tetramers and nucleic acid for an influenza virus HA. In one embodiment, the method further comprises isolating NA and HA from the cell. In one embodiment, the cell is an insect cell. In one embodiment, the cell is a CHO, MDCK, Vero, or EB66 cell.

[0128] Also provided is isolated virus prepared by the above-described methods.

[0129] In one embodiment, a method of immunizing an avian or a mammal is provided comprising: administering to the avian or the mammal a composition having an effective amount of the above-described

virus. In one embodiment, the composition comprises at least one other different influenza virus. In one embodiment, the mammal is a human. In one embodiment, the composition is administered intranasally or via injection.

[0130] Further provided is a method comprising passaging the virus in eggs.

[0131] The invention will be described by the following non-limiting examples.

Example I

[0132] Neuraminidase (NA) is one of the major transmembrane glycoproteins of influenza viruses. It has been suggested that antibodies against NA play important roles in preventing influenza virus infection. However, the current influenza vaccines, which are made by inactivating egg-grown influenza viruses and purifying virus antigen, do not efficiently elicit the production of anti-NA antibodies. One possible reason for the low production of anti-NA antibodies is the structural instability of the NA protein, which functions as a homo-tetramer; the NA tetramer is apparently disrupted during the antigen purification process. Therefore, the amount of NA contained in vaccines is insufficient to elicit the production of anti-NA antibodies.

[0133] The establishment of a method that stabilizes the NA tetrameric structure may solve this problem. The amino acids 48C and 50C in NA have previously been introduced to NA in vitro (Silva et al., 2013); however, the effect of these amino acids on influenza virus replication or replication efficiency was unknown. As described herein, recombinant influenza viruses containing either the NA-48C or NA-50C mutation or both mutations expressed a stabilized NA tetramer and replicated efficiently.

Methods

[0134] A 6+2 reassortant influenza viruses containing the HA and NA gene segments from A/Singapore/GP1908/2015 (H1N1) pdm09 in the backbone of high-yield A/Puerto Rico/8/1934 (H1N1) was prepared using reverse genetics and propagation in hCK cells at 37° C. Mutant viruses with either the NA-T48C or NA-N50C mutation or both mutations were generated. hCK cells were infected with these viruses at a MOI (multiplicity of infection) of 1. Cells were lysed at 9 hours post-infection with or without DTT, and NA was detected by western blotting.

Results

[0135] In the presence of DTT, only the band representing the NA monomer was detected. In the absence of DTT, the band representing the WT-NA dimer was detected. For the mutant viruses, bands representing tetrameric NA-T48C, NA-N50C, and NA-T48C/N50C were detected. This result demonstrates that the ratio of tetrameric NA was increased by NA-T48C, NA-N50C, or both. Stabilization may be detected by any method, e.g., sialidase activity.

Conclusion

[0136] The NA-48C and NA-50C mutations, either singly or in combination, can stabilize the NA tetrameric structure. These amino acid mutations are thus helpful to establish a new vaccine strain that can elicit greater amounts of NA antibodies compared with those elicited by current vaccine strains.

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foregoing specification this invention has been described in relation to certain preferred embodiments thereof, and many details have been set forth for purposes of illustration, it will be apparent to those skilled in the art that the invention is susceptible to additional embodiments and that certain of the details described herein may be varied considerably without departing from the basic principles of the invention.

Claims

1. A method of preparing influenza virus having stabilized NA tetramers or of preparing stabilized NA tetramers, comprising: contacting a cell with a) one or more vectors comprising nucleic acid for an influenza virus neuraminidase (NA) segment encoding a modified NA monomer that forms virions having stabilized NA tetramers relative to a corresponding influenza virus having a parental NA viral segment encoding an unmodified NA monomer, nucleic acid for an influenza virus PA segment, nucleic acid for an influenza virus PB1 segment, nucleic acid for an influenza virus PB2 segment, nucleic acid for an influenza virus NP segment, nucleic acid for an influenza virus NS segment, nucleic acid for an influenza virus M segment, and nucleic acid for an influenza virus HA segment, in an amount effective to produce influenza virus having stabilized NA tetramers; or b) one or more vectors comprising nucleic acid for an influenza virus NA segment encoding a modified NA monomer that forms virions having stabilized NA tetramers relative to a corresponding influenza virus having a parental NA viral segment encoding an unmodified NA monomer, thereby producing a stabilized NA, wherein the modified NA monomer comprises a modified NA stalk that results in the stabilized NA tetramers, wherein the modified NA has a cysteine at position 48 or position 50 or both positions of 48 and 50 relative to the numbering of NA of SEQ ID NO:37.
 2. The method of claim 1a wherein the NA is N1, N2, N3 or N5 or the HA is H1, H3, H5 or H7.
 3. The method of claim 1 wherein the cell is a mammalian cell.
 4. The method of claim 1 wherein the cell is a 293T, PER.C6®, MDCK, MvLu1, CHO or Vero cell.
 5. The method of claim 1 wherein the cell is an avian egg.
 6. The method of claim 1 wherein the cell is a CHO, MDCK, Vero, or EB66, or insect cell.
 7. The method of claim 1a further comprising isolating NA and HA from the cell.
 8. The method of claim 1a further comprising isolating influenza virus from the cell.
 9. The method of claim 1b further comprising isolating NA from the cell.
 10. The method of claim 1 wherein the modified NA further comprises a deletion, an insertion, at least one other amino acid substitution, or any combination thereof.
 11. The method of claim 1 wherein the modified NA has a cysteine at position 48.
 12. The method of claim 1 wherein modified NA has a cysteine at position 50.
 13. The method of claim 1 wherein modified NA has a cysteine at position 48 and position 50.
 14. The method of claim 1 wherein the NA stalk is further modified within residues 1 to 10 or residues 10 to 20 from the C-terminus of the transmembrane domain.
 15. The method of claim 1 wherein the NA is N1, N2, N3 or N5.
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